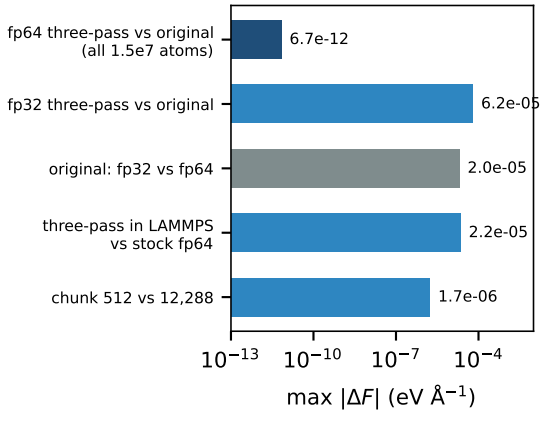
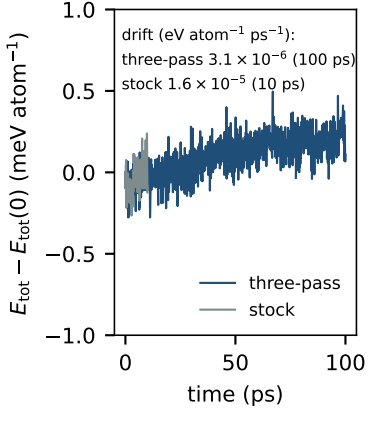
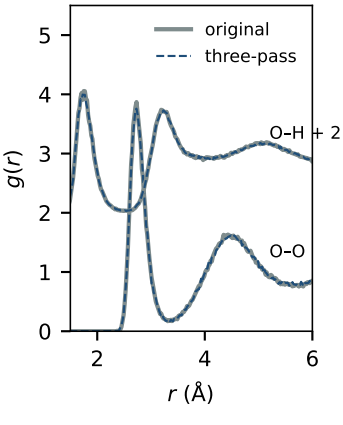

a force deviations
fp64 three-pass vs original
(all 1.5e7 atoms)
6.7e-12
fp32 three-pass vs original
6.2e-05
original: fp32 vs fp64
2.0e-05
three-pass in LAMMPS
vs stock fp64
2.2e-05
chunk 512 vs 12,288
1.7e-06
10^{-13}
10^{-10}
10^{-7}
10^{-4}
max $|\Delta F|$ (eV Å^{-1})


b NVE, 384 atoms, 1 fs
drift (eV atom^{-1} ps^{-1}):
three-pass 3.1×10^{-6} (100 ps)
stock 1.6×10^{-5} (10 ps)
three-pass
stock
$E_{tot} - E_{tot}(0)$ (meV atom^{-1})
time (ps)


c NPT water, 300 K
original
three-pass
O–H + 2
O–O
$g(r)$
r (Å)